

Synopsis

Disease Risk Prediction using Machine Learning Classification

Problem Id: 06

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- **Problem Description:**

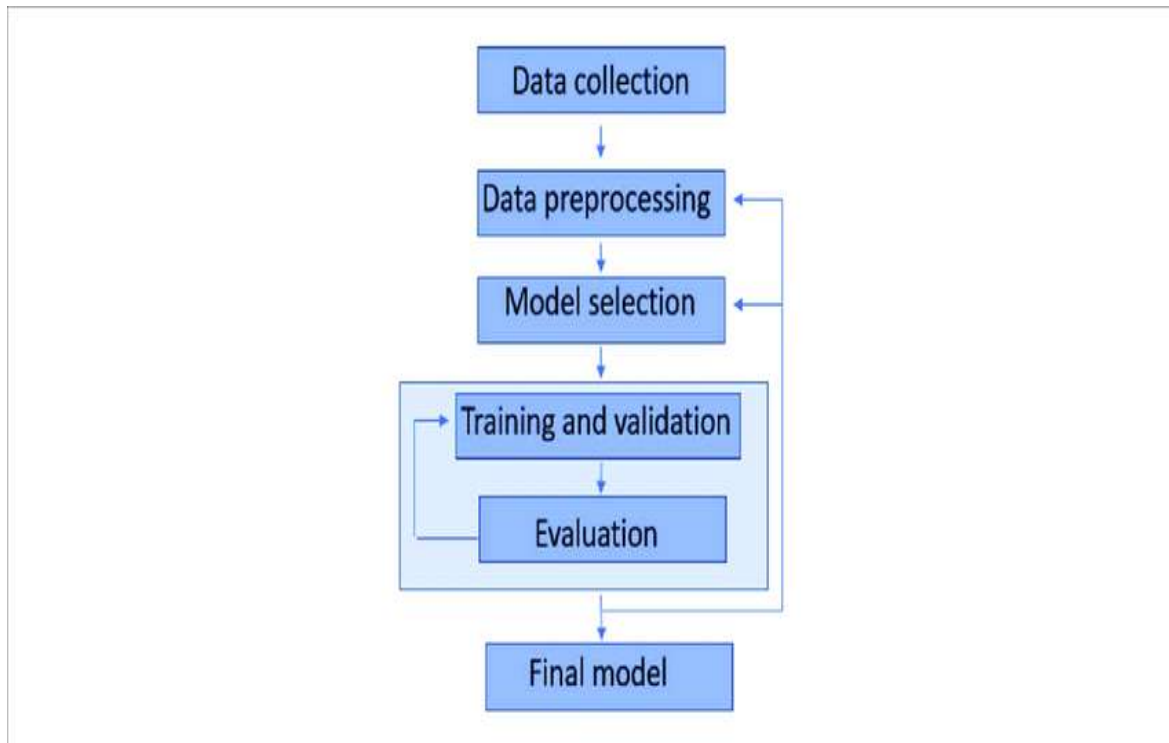
The prediction of disease risk through analysis of patient clinical data establishes the foundation for early diagnosis and preventive medical treatment. The project develops a complete machine learning classification system which uses patient health data to forecast disease risk.

The system starts with the selection of a clinical dataset which undergoes data preprocessing that includes missing value handling and data normalization and data transformation to create high-quality input materials. We employ feature selection methods to determine which medical attributes most significantly impact our analysis. The results improve model performance while making it easier to understand how the models function.

The research tests different classification techniques by using Logistic Regression and K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) methods. The model evaluation uses standard classification indicators which include accuracy and precision and recall and F1-score and confusion matrix.

The project demonstrates a complete machine learning process which transforms raw clinical data into validated predictive models while implementing best machine learning system development practices.

Steps of Implementation:



1. Dataset Curation (Data Collection)

A suitable clinical dataset is selected (e.g., diabetes or heart disease dataset). The dataset contains patient-related features such as age, BMI, blood pressure, glucose levels, etc.

2. Data Cleaning (Preprocessing)

- Handle missing values using mean/median imputation
- Remove duplicate records
- Detect and handle outliers
- Ensure consistency in data formats

Feature Standardization (StandardScaler)

The team uses StandardScaler to normalize data because clinical features contain different measurement systems and International System of Units. The process converts all features into a standardized format which results in a mean value of 0 and a standard deviation of 1. The current step shows its significance because K-Nearest Neighbours (KNN) and Support Vector Machine (SVM) algorithms require proper feature scaling for their operations. Proper standardization improves model performance, convergence speed, and prediction accuracy.

3. Model Building (Classification Models)

Classification models are trained using the processed clinical dataset to predict disease risk. The models used in this project include **Logistic Regression, KNN, and SVM**.

These models learn the relationship between input features such as **age, BMI, blood pressure, glucose level, and insulin levels** and the target variable (disease risk).

4. Model Evaluation(Performance Analysis)

The trained models are evaluated using standard classification metrics to measure their effectiveness. These include **accuracy, precision, recall, and F1-score** to assess prediction quality.

A **confusion matrix** is used to visualize correct and incorrect classifications.

These evaluation techniques help identify the best-performing model and highlight areas for improvement.

- **Expected Modules/ Libraries to be used :**

NumPy:

Used for numerical operations and efficient handling of arrays.

Pandas:

Used for data manipulation, cleaning, and handling structured datasets.

Matplotlib / Seaborn:

Used for data visualization such as confusion matrix, ROC curves, and performance comparison graphs.

Scikit-learn:

Provides machine learning tools including:

- **StandardScaler (for normalization)**
- **Logistic Regression**
- **K-Nearest Neighbours (KNN)**
- **Support Vector Machine (SVM)**
- **Train-Test Split**
- **Evaluation metrics**

- **Expected Evaluation metrics and Plots to be used:**

1.Accuracy

Measures the overall correctness of the model.

2. Precision

Indicates how many predicted positive cases are actually correct.

3 . Recall

Measures how well the model identifies actual positive cases.

4 . F1-Score

Harmonic mean of precision and recall, useful for imbalanced datasets.

5 . Confusion Matrix

Displays true positives, true negatives, false positives, and false negatives.

Required topics from the subject:

Data Structures (Arrays & DataFrames):

Used to store and manage clinical data efficiently using NumPy arrays and Pandas DataFrames.

Functions:

Functions are used to divide the program into smaller, manageable parts. Separate functions can be created for tasks like data collection, preprocessing, scaling using StandardScaler, and training the Linear Regression model.

File Handling:

Used to save and load the dataset (e.g., CSV files) for future use and analysis.

Loops & Conditionals:

Used for iterative data processing, model training, and decision-making operations.

Libraries & Modules:

Used to implement machine learning models and data analysis. NumPy and Pandas handle data, Scikit-learn provides classification models, and Matplotlib/Seaborn are used for visualization.

Platform Required:**Jupyter Notebook:**

Used as the primary environment for coding, testing, and visualization.

Python (Anaconda Distribution):

Provides all required libraries and package management needed for data analysis and machine learning.

- **Books and Link Sources:**

- <https://scikit-learn.org/stable/>
- <https://www.geeksforgeeks.org/machine-learning/>
- <https://www.geeksforgeeks.org/logistic-regression-in-machine-learning/>
- <https://www.geeksforgeeks.org/k-nearest-neighbours/>